

Paul John H. Punzal

Full-Stack Developer | Mobile App Developer

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PROFESSIONAL SUMMARY

IT graduate with experience building real-world systems across web, mobile, and IoT — from a Raspberry Pi-powered gate system with AI helmet detection and OCR (Capstone) to a full SPED school management platform deployed during OJT. Comfortable working across the stack and picking up new tools as needed. Enjoys backend work, particularly *REST API design* and database structuring. Interested in AI and automation, with hands-on experience using YOLO, OCR, and locally-hosted LLMs for personal exploration.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Dart, Python, PHP, C#

Frameworks & Libraries: React (Web), React Native (Expo), Next.js, Laravel, Flutter, Firebase SDK, YOLO (Object Detection), and NestJS

Databases: MySQL, SQLite, Firebase Firestore, Prisma

Tools & Platforms: Git, GitHub, Figma, Android Debug Bridge (ADB), Unity, Blender, Ollama (LLaMA 3 — exploration)

TECHNICAL PROJECT EXPERIENCE

MotorPass — Smart Motorcycle Gate Pass System

(Capstone Project)

Role: Full-Stack Developer | Stack: Python, Raspberry Pi 4, Firebase, SQLite, YOLO/ONNX, OCR, Biometrics

- Engineered an IoT-based automated campus entry system on a Raspberry Pi 4, integrating hardware and cloud services to replace manual gate pass workflows and improve throughput.
- Developed a custom-trained YOLO/ONNX machine learning model for real-time helmet detection, automatically enforcing campus safety compliance without human intervention.
- Implemented multi-factor authentication combining OCR-powered driver's license scanning with fingerprint biometrics to ensure secure and verified vehicle entry.
- Architected a dual-database strategy using SQLite for resilient local event logging and Firebase Realtime Database for live synchronization, providing administrators a live monitoring web dashboard with zero data loss during connectivity drops.

Little Lion — SPED School Management System

(During OJT)

Role: Lead on Developer & UI/UX | Stack: JavaScript, React, Firebase Firestore, Firebase Auth, CSS

- Led end-to-end development of a specialized administrative platform for a Special Education school during OJT deployment, covering requirements gathering, architecture, and delivery.

- Architected a role-based access control (RBAC) system providing five distinct, tailored portals for Admins, Staff, Teachers, Therapists, and Parents—ensuring each stakeholder only accesses relevant data.
- Designed accessible and calming frontend interfaces using intentional color palettes (greens and whites) and clear information hierarchies to maximize usability for all user groups, including parents with varying technical literacy.
- Built a real-time parent communication hub featuring a structured "Concerns" ticketing system and live daily activity digests, significantly improving transparency and parent-teacher engagement.

E-Learning & Assessment Platform

(Freelance)

Role: Full-Stack & Mobile Developer | Stack: Flutter, Dart, PHP (Vanilla), MySQL, REST APIs

- Developed a cross-platform mobile e-learning application using Flutter/Dart, paired with a custom Vanilla PHP backend to deliver a fully self-contained educational ecosystem.
- Engineered a dynamic quiz and assessment engine supporting score tracking, attempt history, and randomized difficulty levels to provide adaptive learning experiences across multiple lesson modules.
- Designed and implemented RESTful APIs to manage complex relational data flows between the mobile frontend and MySQL database, ensuring clean data contracts and scalable integration.
- Executed major database schema migrations and role restructuring to optimize data flow and support new feature rollouts without downtime, while implementing secure OTP-based user authentication.

Offline Grocery List — Smart Utility App

Role: Mobile Developer | Stack: React Native (Expo), TypeScript, Local Storage APIs, OCR

- Developed an offline-first mobile productivity application in TypeScript, architected to function entirely without internet connectivity by leveraging on-device local storage APIs.
- Integrated device-native camera access with AI-powered OCR to enable users to instantly capture and parse product information directly from packaging, eliminating manual data entry.
- Implemented on-device scheduling algorithms and local notification management to handle list reminders and session persistence, ensuring a seamless and uninterrupted user experience.

EDUCATION

Bachelor of Science in Information Technology

Pambayang Dalubhasaan ng Marilao | Graduated: May 2026

SEMINARS

- IT Project Management, 2024
- Cyber Security, 2025
- Intelligent Design Converging Architecture, Security & AI, 2025
- Understanding Agile Development: From Sprints and Roles to Code Reviews and CI/CD, 2025
- Beyond Coding: Exploring Business and Product Roles in the Tech Industry, 2025